

★ Skills

Data Analysis:

- Jupyter Notebooks
- SQL
- Pandas
- Tableau
- Spreadsheets
- Relational databases
- Google BigQuery
- Google Cloud Dataflow
- Microsoft Excel
- Microsoft Access
- Microsoft PowerPoint
- Microsoft Power BI
- Microsoft Power Query
- Artificial Intelligence (AI)

Web Development:

- JavaScript
- ECMAScript
- Node.js
- Express.js
- Python
- Flask
- Git
- Github
- HTML
- CSS
- Linux
- JSON
- REST
- RESTful APIs
- Postman
- MySQL
- PostgreSQL
- NoSQL
- MongoDB
- Docker
- Object-Oriented Programming
- Unit Testing
- Design Patterns
- Google Cloud Platform
- Agile (Scrum)

🔗 Language Skills

• English

- Listening: C2
- Reading: C2
- Spoken production: C2
- Spoken interaction: C2
- Writing: C1

• Japanese

- Listening: B1

Mohammad Sheharyar

| Gender: Male Nationality: Pakistani | Mobile phone: +92 3355965510 | Email address: stormbrewer@yandex.com | Website: <https://mOs111.github.io/> | LinkedIn: <https://www.linkedin.com/in/mOs111/>

👤 About me

Practicing Data Analyst with a Software Engineering background. Experienced in SQL and Pandas for data wrangling and analysis in BigQuery, Jupyter, Kaggle, or Colab Notebooks. Possesses knowledge of R's Tidyverse package. Utilizes Power BI and Tableau to create industry-standard visualizations and dashboards for communicating insights dynamically. Addresses Business Intelligence processes with spreadsheets (Microsoft Excel and Google Sheets). Comfortable with Machine Learning Deployment, and Back-End Development. Proficient in Python and Node.js for server-side logic and database management. Strong understanding of Google Cloud Platform (GCP). Familiar with integration through ELT/ETL pipelines and processes.

💼 Work experience

Software engineer

01/2025 – 04/2025

IDEVNERDS | Lahore, Pakistan

Back-End Web Dev Intern

Built complete RESTful APIs with Express.js for an e-commerce platform

Optimized APIs using async processing (non-blocking) and leveraging connection pooling with ORM/ODM

Wrote Middleware to implement JSON Web Token Authentication in cookies for separate client and admin routes

Wrote Middleware to validate data using the Joi library before insertion into databases

Encrypted data with the bcrypt library

Created Dockerfiles for deploying containerized databases

Used Docker CLI for managing images and containers

Used Sequelize ORM and CLI to interface with MySQL and PostgreSQL databases

Used mysql and psql CLIs, MySQL Workbench & PGAdmin

Used Mongoose ODM to interface with MongoDB databases

Used the mongosh CLI & MongoDB Compass

Tested APIs with curl and Postman (VS Code extension)

Managed project packages using Node Package Manager (npm)

Business or sector Information and communication | **Website:** <https://idevnerds.com/>

Software engineer

05/2024 – 11/2024

MachKnowSoft | Abbottabad, Pakistan

Software Development Intern

Created Python application software

Created GUIs using the Python tkinter library

Created websites/web apps using Flask micro-framework

Wrote test cases and unit tests for apps

Created scripts for Linux Bash using Python os, shutil, and psutil modules

Managed project packages using Python Package Manager (pip)

Business or sector Information and communication | **Website:** <https://www.machknowsoft.net/>

+ Certificates

CompTIA Data+

06/08/2026 – 06/08/2029

https://www.credly.com/badges/9e6e725c-b983-475d-888f-c18cdeda3230/linked_in_profile

- Reading: B1
- Spoken production: B1
- Spoken interaction: B1
- Writing: A2

Google Data Analytics

<https://coursera.org/verify/professional-cert/3ECBW1QJD9N6>

Google IT Automation with Python

<https://coursera.org/verify/professional-cert/K5LZWFKPHB9F>

Google Project Management

<https://coursera.org/verify/professional-cert/L5XJ3J2MTCPL>

UC San Diego ALGS200x: Algorithmic Design and Techniques

<https://courses.edx.org/certificates/778698a318064eb2b48c950684885c9c>

UC San Diego ALGS201x: Data Structures Fundamentals

<https://courses.edx.org/certificates/df74a80d90e74d1a80c4b839737748e6>

■ Projects

Social Media Addiction Exploratory Data Analysis

Used Python, Pandas & Matplotlib to check, clean, analyze, and visualize an open, high-quality dataset sourced on Kaggle

Created hypotheses to test

Produced statistics for the dataset

Ensured the statistical significance of the analysis

Delivered insights on how Social Media addiction impacts young adults globally using Matplotlib plots

<https://www.kaggle.com/code/mohammadsheharyar/social-media-addiction-eda>

FDA Food Events Exploratory Data Analysis

Used SQL with the Google BigQuery API to check, clean, analyze, and visualize an open, high-quality dataset sourced from BigQuery public datasets

Utilized Window Functions, aggregations, subqueries, JOINS, and temporary tables

Analyzed the number of serious health events caused by the food industries

Analyzed which food categories severely affected infants under the age of 3

Plotted the data of affected infants

Analyzed the number of fatal outcomes over 25 years

<https://www.kaggle.com/code/mohammadsheharyar/fda-food-events-eda>

Gen AI course Capstone Project

Utilized Google Gen AI Python SDK to develop a solution for AI-generated Subtitles for the Deaf and Hard of Hearing (SDH) in a structured JSON format with few-shot prompting

Interpreted American Sign Language (ASL) using the grounding feature to ensure accuracy instead of a Retrieval Augmented Generation (RAG) solution

<https://www.kaggle.com/code/mohammadsheharyar/sdh-generation-with-google-gen-ai>

Data Analysis Project

Used the Pandas and NumPy libraries to generate and then simulate the analysis of Social Media post categories

Cleaned data with dataframe methods

Used dataframe methods and Matplotlib plots to visualize and analyze data

<https://github.com/MOS111/Coursera-DataAnalysis-Project/blob/main/SocialMediaDataAnalysis.ipynb>

World Bank/UNWTO Tourism data (1995–2020)

Sourced open data from the World Bank data bank for national receipts from and expenditure on tourism

Retained original data in separate sheets

Sorted, filtered, and applied conditional formatting on the data to clean it of missing values

Created pivot tables to consolidate relevant data for 5 years (2016–2020)

Created basic bar graph visualization for receipts and expenditures

https://docs.google.com/spreadsheets/d/1bVY-9Gn5vqugJRcGC_jtQwGvKOKWA3NCS_TJ89kQ7-g/edit?usp=sharing

Visualization of the World Bank/UNWTO Tourism data (2020)

Cleaned and formatted the World Bank/UNWTO open dataset in the spreadsheets mentioned above

Visualized the receipts in millions of dollars for international tourism during 2020 as a heatmap

Created an accompanying bar graph of the same

<https://public.tableau.com/app/profile/mohammad.sheharyar/viz/WorldBankUNWTOTourism2016-2020/Sheet2>

Visualization of Movies Performance

Created a dashboard of the profitability of movie genres during 2 years (2014–2016)

Visualized genre profitability over time as a line graph

Showed the revenue earned by famous actors as a circle packing diagram

Charted the number of movies actors acted in per genre as a stacked bar graph

<https://public.tableau.com/app/profile/mohammad.sheharyar/viz/MoviesFinancialPerformance2014-2016/Dashboard1>

Personal Website

Used Synchronized Multimedia Integration Language (SMIL) with HTML SVG elements to create an unorthodox navigation menu

Used jQuery animate() and css() methods with CSS keyframe animations to create a neon glow effect

Created a simple loop-around carousel to display certificates

Called an external jokes API using JavaScript Fetch API to serve programming jokes from JSON payloads

Shortened page loading time with lazy loading of images

<https://mOs111.github.io/>

Recommendation System (University Capstone)

Created a hybrid recommendation system for Western movies and deployed it as a single-page application (SPA)

Implemented the system as a Machine Learning model using the k-Nearest Neighbors algorithm using the scikit-learn library

Stored the model as a serialized Pickle file

Ran the model as a Flask app on a WSGI server

https://github.com/MOS111/Rec-Sys/blob/main/Rec_Sys.ipynb
<http://msheharyar111.pythonanywhere.com/>

🎓 Education & Training

Bachelor of Science

09/2018 – 09/2023

Virtual University of Pakistan | Lahore, Pakistan

Field(s) of study: Software and applications development and analysis | **Final grade:** 3.65 |
Level in EQF: 6 | **Website:** <https://vu.edu.pk>